

WaveafterWave Simulator

System Architecture Overview

Core Architecture

The WaveafterWave simulator is built as a single-page web application using JavaScript with the Web Audio API. The system follows a modular architecture with these key components:

1. **Audio Engine** (WaveafterWave class)
 - Manages Web Audio Context and master gain chain
 - Handles ambient sound generation and contextual effects
 - Processes notification sounds with priority-based scheduling
2. **Event Processing System**
 - PriorityQueue for handling simultaneous notifications
 - JSONNotifier for loading and scheduling JSON event streams
 - Real-time event filtering and spatial audio positioning
3. **Context Management**
 - Four environmental contexts with distinct audio profiles
 - Dynamic audio processing chain per context
 - Real-time context switching with smooth transitions

Audio Processing Chain

Ambient Source → Ambient Gain → [Stereo Spread] → [Delay] → [Reverb] → Filter → Master Gain → Output

How to Experience Different Scenarios: *Please note these might take some time before you hear the sounds.*

1. Context Exploration

Office Environment:

- Select "Office" context
- Listen for quiet office and system hum with occasional keyboard typing
- Notice the low-pass filtering creating a professional atmosphere

Street Environment:

- Select "Street" context
- Experience traffic noise with wind effects
- Observe band-pass filtering for safety awareness
- Notifications are more centered for environmental awareness

Café Environment:

- Select "Café" context
- Hear crowd murmur with cup clinking sounds
- Experience stereo spread and reverb for spaciousness
- Notice how notifications blend with social atmosphere

Home Environment:

- Select "Home" context
- Listen to subtle fridge drone with minimal filtering
- Experience the most natural, unobtrusive background

2. Advanced Features Demonstration

Simultaneous Event Handling:

- The system processes up to 3 events simultaneously
- High-priority events (≥ 3) trigger text-to-speech
- Spatial positioning prevents auditory masking:
 - Tweets: Left-panned (-0.7)
 - Emails: Left-center (-0.3)
 - Texts: Center (0)
 - Calls: Right-center (0.3)
 - Voicemails: Right-panned (0.7)

Priority-Based Scheduling:

- Events are processed in priority order (1-4, with 4 being highest)
- High-priority events receive volume boosts and distortion effects
- Rapid low-priority bursts are summarized to prevent overload

Sonification Scheme Recap

Notification Sounds

- **Tweets:** Bright FM-modulated chirps (800Hz sine, 0.1s)
- **Emails:** Soft filtered chimes (600Hz sine, 0.3s)
- **Texts:** Quick percussive pings (1000Hz square, 0.05s)
- **Calls:** Urgent ringing tones (400Hz sawtooth, 0.5s)
- **Voicemails:** Record-click followed by TTS (300Hz triangle, 0.4s)

Contextual Adaptation

Each context modifies both ambient sounds and notification presentation:

- **Volume Adjustment:** Context-appropriate loudness levels
- **Filtering:** Frequency shaping for environmental suitability
- **Spatial Processing:** Safety-aware panning in street contexts
- **Effects:** Reverb and delay for environmental realism

Parameter Mapping

- **Priority → Volume & Brightness:** Louder/brighter = more urgent
- **Emotional Valence → Harmony:** Consonant = positive, dissonant = negative
- **Information Density → Rhythm:** Frequent events summarized via tempo
- **Sender Identity → TTS Voice:** Distinct timbres aid recall

Changes from the Original Design

1. Simplified Audio Effects Chain

- **Original:** Complex multi-stage processing with convolution reverb
- **Implemented:** Streamlined effects with fallback mechanisms
- **Reason:** Substituting non playable sounds or audio load errors

2. Added Debug Features

- **Original:** No debugging tools
- **Implemented:** Audio state monitoring and test tones
- **Reason:** Development and troubleshooting needs

Not Implemented: Advanced Contextual Adaptation Features

The following contextual adaptations from the original design document were not implemented in the current version:

- Working Out Context: High-frequency boost and TTS reduction
- Walking Context: Dedicated high-pass filtering and spatial offset
- Socializing Context: Volume automation and brief motifs
- Presenting Context: Priority 4-only pulse notifications

Reason: These features were too advanced to implement.

Additional Features Beyond Requirements

1. Song Integration

- Dedicated "Waves" by Mr. Probz playback
- Automatic simulation pausing during song playback
- YouTube integration for original song access and credit

2. Audio Controls

- Real-time master volume slider
- Audio context state monitoring
- One-click audio testing functionality
- Visual feedback for all audio states